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A DEEP-LEARNING CNN APPROACH TO TORNADO PROBABILITY PREDICTION USING NEXRAD RADAR DATA

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Abstract

The United States National Weather Service maintains a fleet of NEXRAD weather radars. These radars produce new scans of the atmosphere every 3-7 minutes, which provide a rich description of the state of precipitation in areas surrounding the radar. An important use for NEXRAD radar data is for the identification of tornadic features within the data, such as mesocyclones and hook echoes. This is an important tool for the creation of early warnings for any potential impacts on civilians in the path of the storms. This paper introduces MARTAN - a deep-learning CNN model that utilises the TorNet dataset to formulate an estimation of ongoing tornado likelihood within a given sector of the radar data. The model ingests four dual-tilt radar moments for its analysis; Reflectivity (dBZ), Velocity (v), Spectrum Width (σ_v) and Correlation Coefficient (ρ_{hv}). The results strongly indicate that CNNs can effectively generalise and analyse tornadic representation within radar data, and produce sensible predictions of tornado likelihood.

Keywords: CNN; Deep-Learning; Tornado Prediction; Weather Radar; Machine Learning; NEXRAD

1. Introduction

Recent years have seen great advancement in the accuracy and variety of radar-based tornado detection algorithms. The purpose of these algorithms is to extract features from NEXRAD radar data, and perform some form of analysis upon the data to produce an accurate prediction of the chance of an ongoing tornado at the location. More recent algorithms typically employ a machine learning component to help formulate these predictions, however, most of the recent solutions typically use random forests on extracted feature values [1], rather than using deep learning architectures. Whilst this method brought a great leap in tornado detection accuracy compared to the previous purely algorithmic solutions [2], extracting features in this way loses a lot of contextual information from the data, such as storm presentation and larger-scale features such as the presence and intensity of the rear-flank downdraught or inflow.

This paper introduces MARTAN (Mesoscale And Radar Tornado Analysis Network), a suite of three machine-learning models that predominantly use CNNs (Convolutional Neural Networks) to analyse segments of radar sweeps, in an attempt to form an accurate prediction for the likelihood of ongoing tornadoes using deep learning. Novel contributions are discussed, such as the presentation of the new multimodal architecture, in addition to a comprehensive explainable AI analysis, including CAM and ablation studies of the differing radar moments and tilts. By striving to improve the accuracy of tornado detection algorithms, now-casters will have access to better tools to assist with their tornado warning issuance process, which could lead to fewer false positives and earlier warning issuance.

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This would increase confidence in the relevant weather authority, and potentially give at-risk civilians more time to prepare for any impending danger. The results of this study suggest that MARTAN can identify impending tornadogenesis effectively and with notable skill.

2. Methods

This section describes the process by which the dataset was adapted from the original work by Veilette et al. [3]. Samples were taken from TorNet and further refined to better suit MARTAN's specifications. HRRR analysis variables are extracted and linked to the samples, and the data pre-processing methods are explained. The model's architecture is outlined, and architectural decisions are reinforced. The model training methodology is also provided.

2.1. Dataset

The original dataset used to train the models in this paper is provided by Veilette et al. [3]. This dataset provides pre-processed segments of radar data, which are weakly labelled with information such as latitude and longitude, sample type (tornado, warned, null), and an EF rating, should the sample contain a tornado. The velocity data comes de-aliased using the method also described in Veilette et al's paper [3]. This dataset contains 203,133 samples, collected from events identified by the SPC's storm event database between August 31st 2013 and the September 30th 2022 [13]. Of the 203,133 total samples within the dataset, 13,857 (6.8%) depict positive "tornado" samples, 64,510 (31.8%) depict "warned" samples, and 124,766 (61.4%) depict "random" samples. "Tornado" samples verifiably depict ongoing tornadoes, as each event is linked with an on-site survey completed by the National Weather Service (NWS). Warned samples depict tornadic features significant enough on radar for NWS now-casters to issue a tornado warning, without a tornado being confirmed by an on-site survey. Random samples are procured from a wide range of meteorological systems, and their collection is described in detail within the original paper [3].

A list of all tornado and thunderstorm warnings was collected from the Iowa Environmental Mesonet (IEM) database between August 31st 2013, and September 30th 2022 [5]. Each TorNet sample is tested for spatial and temporal overlap with the IEM warnings database. If the sample is labelled as either tornadic or warned, yet an overlapping warning could not be found, the sample is removed from the dataset. If the sample is labelled "random" and a warning overlap is discovered, this sample is placed in a new category "severe", which represents severe thunderstorms that should not depict tornadic features. Samples labelled "random" without a linked warning are placed into a different category labelled "negative", which represents clear negative samples. If the sample has a linked warning, a more accurate centroid latitude and longitude are extracted from the warning's polygon. If the sample does not have a linked warning, the TorNet sample's latitude and longitude metadata are used as the sample's centroid instead. This process creates a more accurate and descriptive dataset, at the cost of cutting a portion of the original dataset. After this process, the new dataset contains 178,246 samples in total, of which 12,102 (6.8%) are labelled "tornado", 48,785 (27.4%) are labelled "warned", 30,540 (17.1%) are labelled "severe", and 86,819 (48.7%) are labelled "negative".

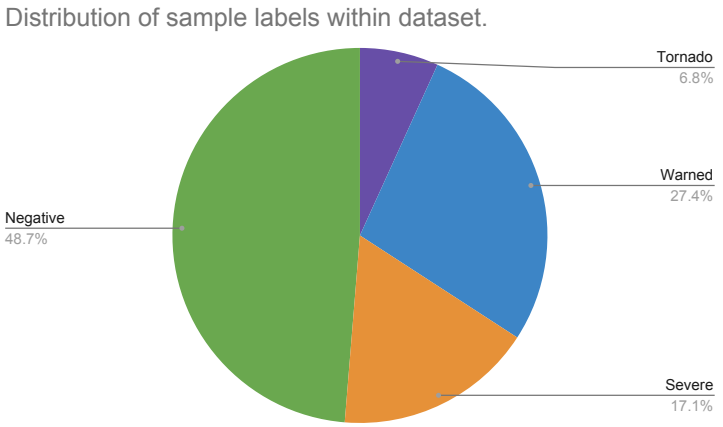


Figure 1. Final dataset composition

The centroid of each sample is used to query the HRRR ZARR repository maintained by the MesoWest group at the University of Utah [4]. For each sample, the HRRR analysis data was extracted from the first frame of the most recent HRRR model run. Nine descriptive environmental variables were extracted from the HRRR analysis and linked to each sample (*lcl_height*, *dewpoint*, *mlcape*, *cape_0_3km*, *srh_0_1km*, *shear_0_1km_U*, *shear_0_1km_V*, *shear_0_6km_U*, *shear_0_6km_V*). Overall shear magnitude at the 1km and 6km levels are derived from the sample's U and V components, according to the formula $S = \sqrt{S_u^2 + S_v^2}$. These variables are linked to the dual-tilt radar data, in addition to the sample's label, EF number, timestamp, latitude, and longitude, to form the complete sample file. Each sample is saved in .npz format.

TensorFlow DataSets (TFDS) is used to convert the samples into a format more efficient I/O operations. MARTAN-1's dataset contains the complete dual-tilt radar data for Reflectivity, Velocity, Correlation Coefficient, and Spectrum Width, in addition to the sample's label. MARTAN-1 uses every sample in the new dataset. The dataset is separated into training and testing splits, with 80% of samples entering the training split, and 20% entering the testing split. To ensure equal representation of tornado strength within both splits, the "tornado" samples are separated by their EF rating, with each set of EF ratings separated into the 80% and 20% batches. The samples are then placed into the training and testing splits.

2.2. Pre-processing and data sources

Four radar moments are used across the MARTAN models; Reflectivity, Velocity, Correlation Coefficient, and Spectrum Width. The primary purpose of these moments for tornado detection is as follows:

- **Reflectivity -** Measures the intensity of precipitation at a point. Low values indicate areas of low to no precipitation, whereas higher values could represent intense precipitation, large hail, or potentially lofted debris from tornados in extreme cases.
- **Velocity -** Measures the intensity and direction of the wind at a point. Values are either positive or negative depending on the direction of the object the radar echo returns from. A large gradient between two neighbouring pixels can be an indicator of rotation, and is the primary data type used to identify tornados from weather radar.
- **Correlation Coefficient -** A dual-polarised moment, which measures the uniformity of the size of objects in which the radar echo returned from at a point. High correlation indicates uniformity within an area, which could represent standard rainfall. Low correlation indicates non-uniformity in the size of objects at a location, which could be another indicator of lofted debris.

- **Spectrum Width -** Measures the range of different velocities encountered at a location. A larger spectrum width indicates turbulent wind at a location, whereas a lower spectrum width indicates uniform flow.

The two lowest sweeps are taken for each of these moments, to give the model a limited three-dimensional understanding of the atmosphere. The tilts used are typically at 0.5° and 0.9°, however these vary slightly per radar.

Minimal pre-processing is applied to the radar moments before model training and inference. Velocity data in particular is prone to contamination from several sources, such as range folding and interference from non-meteorological entities. While the velocity data within the TorNet dataset has been dealised, which will reduce contamination from range folding errors in many cases, it is not unusual for large chunks of velocity data to be missing from samples. A Gaussian smoothing algorithm is applied to the velocity data to help fill in missing data with sensible, accurate values.

$$Y(i,j) = (X * G_\sigma)(i,j) = \sum_{m=-k}^{k} \sum_{n=-k}^{k} X(i-m, j-n) \frac{1}{2\pi\sigma^2} \exp\left(-\frac{m^2+n^2}{2\sigma^2}\right)$$

Normalisation is applied to each radar moment during pre-processing. A dataset-wide maximum, minimum, mean and standard deviation are taken for each moment across each tilt. Each tensor is clamped to these maximum and minimum values, which are then normalised between 0,1 for Reflectivity (R), Spectrum Width (S) and Correlation Coefficient (C), and -1,1 for Velocity (V) to account for directional differences. Each moment is linearly interpolated for normalisation, except from Spectrum Width which is logarithmic, so the log1p is taken for each value in the tensor, which is then normalised according to the dataset-wide mean and standard deviation.

$$(R \mid C)_{0\text{–}1}(i,j) = \frac{(R \mid C)(i,j)}{(R \mid C)_{\max}}$$

$$V_{-1\text{–}1}(i,j) = \frac{V(i,j)}{V_{\max}}$$

$$S_{\text{lognorm}}(i,j) = \frac{\log S(i,j) - S_\mu}{S_\sigma}$$

The predominant focus of MARTAN-0 is to identify Areas Of Interest (AOIs) within complete radar sweeps. As the TorNet dataset contains weak labelling (positive/negative) rather than pixel-wise segmentation, weakly supervised object localisation (WSOL) has been utilised to locate these AOIs within the input space [6]. The model learns from positive samples (warned or tornado confirmed) which features generally illicit a tornado warning from professional NWS nowcasters. A Class Activation Mapping (CAM) is then extracted from the final convolutional layer to form a "heatmap" of the pixels that the model considered of highest value when forming its conclusion. A threshold of 80% is applied to each pixel's activation to create distinct AOIs within the input space. A weighted centroid can be calculated from these AOIs to give the centre of mass x,y coordinate, which can be used as a method to initialise both MARTAN-1 and MARTAN-2. GRADCAM++ is used over typical GRADCAM for creating the activation maps, as it has been shown to significantly outperform its predecessor with localisation, especially with identifying multiple AOIs within the input [7].

MARTAN-0 only considers Reflectivity and Velocity data to keep it as light-weight as possible. As mesocyclones are a feature within Velocity data, this moment is the most imperative to use. Reflectivity data also helps the model identify storm shape and provide context to the Velocity data.

MARTAN-1 uses the entire set of radar data in an attempt to form a conclusive prediction on whether a radar scan features ongoing tornadogenesis. The model is designed to analyse radar patches of 120 azimuth gates by 240 range bins, as is the native resolution of the TorNet samples. The lowest two tilts of Reflectivity, Velocity, Correlation Coefficient, and Spectrum Width are all used by this model.

2.3. Model architecture

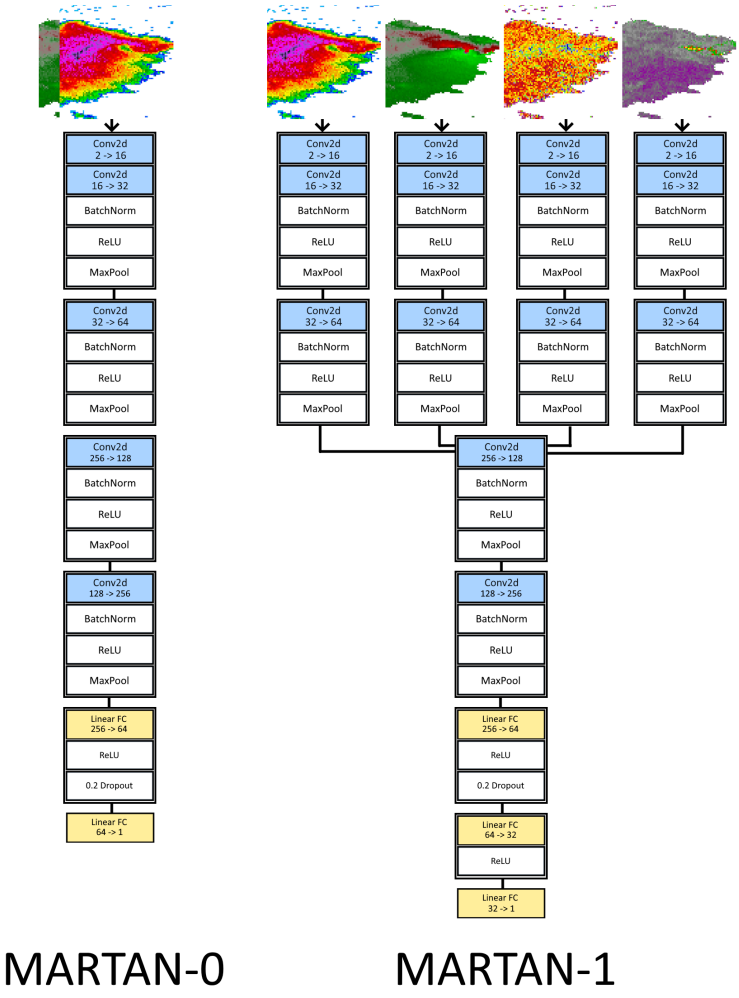


Figure 2. MARTAN-0 and MARTAN-1 model architectures

MARTAN-1 features a separate input head for each radar moment, each containing two convolutional blocks. The first convolutional block employs the double stacked convolutional layer described in ResNet [8] for improved feature identification at a higher resolution, which is important since some important features like velocity gates can appear as a gradient between two individual pixels. Each convolutional layer has a stride of 1, a kernel size of 3, and a padding of 1. The convolutional blocks employ batch-wise normalisation and max pooling after the convolutional layers. After the compartmentalised processing of each moment, the activation maps are concatenated and passed through two further shared convolutional blocks. The feature maps are then flattened and passed through two fully connected layers, before reaching the final prediction logit.

Prior investigations revealed that a multi-input network compared to a more traditional CNN architecture improved both the stability of training, and achieved convergence in fewer epochs, at the cost of a longer training time per epoch. By separating the moments at input, the relationships between both tilts of the same moment can be learned more effi-

ciently before their shared feature map are considered in tandem with the other moments deeper in the network (see Figure 3).

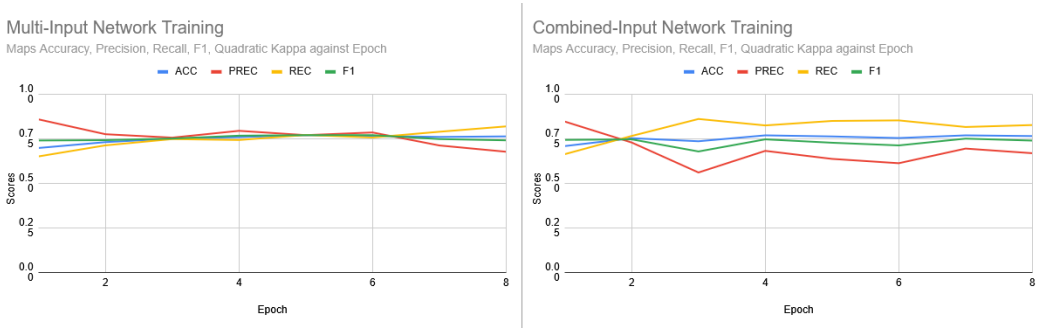


Figure 3. An earlier comparison between multi-input and combined-input tornado detection models.

Coordconv [9] layers were considered over traditional conv2d layers in the convolutional blocks. By running the samples through MARTAN-0 and centring the input tensor around the derived centroid would have, in theory, allowed for a better spatial consistency amongst samples; however in practice, it was discovered that adding the extra coordinate channels added significant confusion to the network (see Figure 4). As a result, these layers were replaced with the standard torch 2d convolutional layers. Other architectural decisions, such as input/output channels, network depth, and pooling type, were investigated by training several test models, and these hyperparameters were set to their optimal values.

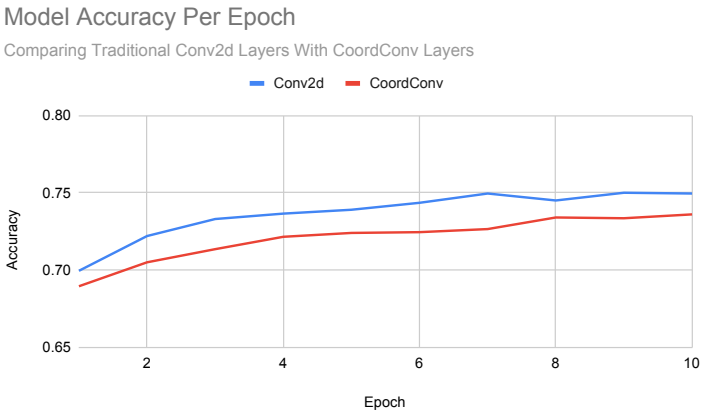


Figure 4. A comparison of training accuracy when using Conv2d and CoordConv layers with MARTAN-1.

2.4. Training

Table 1. Hyperparameters used throughout training.

Hyperparameter	Value
Learning Rate	0.001
TFDS Build Seed	33
K-Fold Seed	3
Positive Prediction Threshold	0.5
Batch Size	32
Workers	2
Positive Sample Weight	13.7058824
Leniency Forgiveness	0.5

Table 2. Hardware specificaftions.

Specification	Component
OS	Ubuntu 25.10
CPU	AMD Ryzen 7 8845HS @ 5GHz
RAM	16GB DDR5 @ 6000MHz
GPU	NVIDIA RTX 4050
VRAM	6 GB
MARTAN-1 Training Time (per Epoch)	65m

As mentioned earlier, the dataset used to train MARTAN-1 consisted of every available sample within the adapted TorNet dataset. As positive samples account for only 6.8% of all samples within the dataset, methods were employed to counteract this extreme class imbalance. Predominantly, a weight was applied to the loss function to boost the loss score for positive samples by the proportion of negative samples to positive samples, calculated as $W_1 = \frac{1.0-0.068}{0.068}$. A method to scale weights proposed by Veilette et al.[3] was also found to have a positive impact on the overall classification accuracy of the model (see figure 5). This method involved reducing the loss for a sample by half if the model's prediction was positive, but the sample was a negative but warned sample. The logic for this is that the model should be penalised less for predicting positive on a sample that elicited a tornado warning from a professional forecaster, without a confirmed tornado, compared to predicting positive on a sample that contains no tornadic features at all.

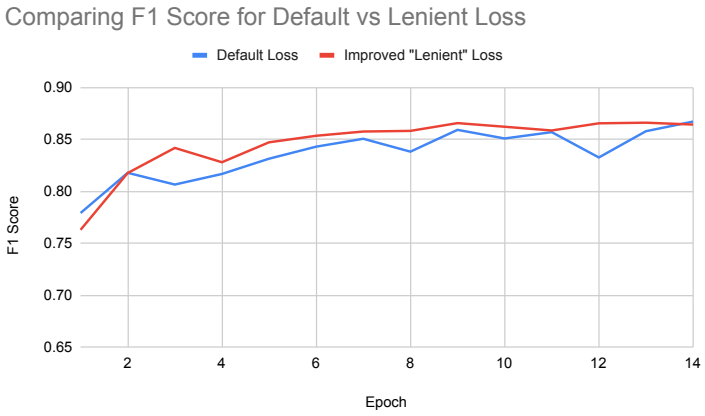


Figure 5. A comparison of training F1 scores for the unchanged loss function vs the new "lenient" loss method. This shows that lenient loss helps the model learn significantly faster in early epochs.

The model was considered to have converged once there was no improvement in log loss after three epochs. After investigation and comparison with alternative options, binary cross-entropy was selected as the loss function, and AdamW was selected as the optimiser. These are both standard choices for binary classification networks.

MARTAN-2 differs from MARTAN-0 and MARTAN-1, as it is not a deep learning model. MARTAN-2 employs a random-forest architecture which has been proven to be very effective in generalising well in weather-related scenarios [1]. The mesoscale variables are used here to estimate the potential for stronger tornados, based on these environmental factors. This model is very experimental compared to the other models, and further details on its suitability is provided in the discussion. Random-forest models construct many binary decision trees sequentially to form a classification prediction, where each tree "votes" for its choice [10]. The average of every tree's prediction forms the overall prediction for the model.

Stratified K-folding was used to separate the dataset into five splits. Stratified sampling is imperative due to the class imbalance, this ensures an equal representation of each class across the splits. The model is trained and tested five times using a different split as the testing split each time, with the evaluation metrics calculated as an average of the metrics across all five splits. The number of predictors (trees) used is a tunable hyperparameter, with 200 predictors discovered to be the optimal value (see Table 3). The final operational model is trained on the entire dataset.

Table 3. F1 scores produced by a random forest model trained on the dataset.

Predictors	Macro F1 Scores
20	0.8034
50	0.812
100	0.8122
200	**0.813**
300	0.8124
400	0.8127
500	0.8122
600	0.8127

3. Results

3.1. MARTAN-0

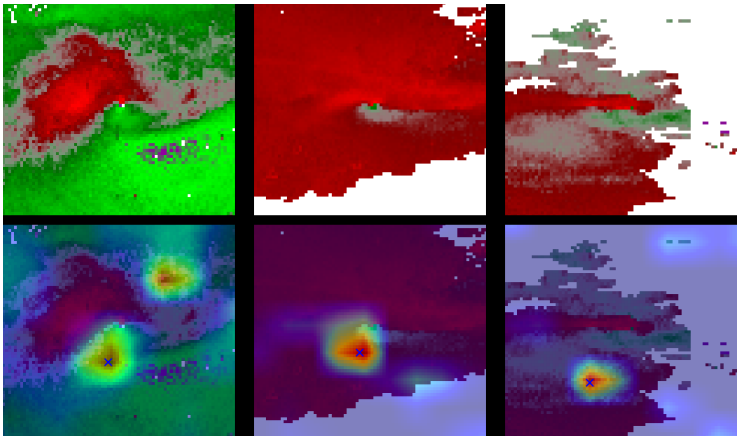


Figure 6. Examples of MARTAN-0 WSOL performance zoomed in on mesocyclones. Inspection shows mesoscale localisation with modest accuracy. The model generally gets in the right area, but shows an interesting preference for the rear-flank feature.

MARTAN-0 attempts to identify AOIs within the radar input to act as an initialiser for downstream models. Figure 6 indicates that the model can localise AOIs within the input space with skill, using only lowest tilt level 2 reflectivity and velocity moments. While creating less precise localisations, earlier epochs identified mesocyclones more accurately than later epochs, suggesting that this WSOL method quickly becomes prone to overfitting, as a noticeable centroid drift occurred within later epochs. This appears to be because the model started placing higher precedence on the rear-flank downdraught feature of cells than the mesocyclone itself, leading to higher activations for that feature.

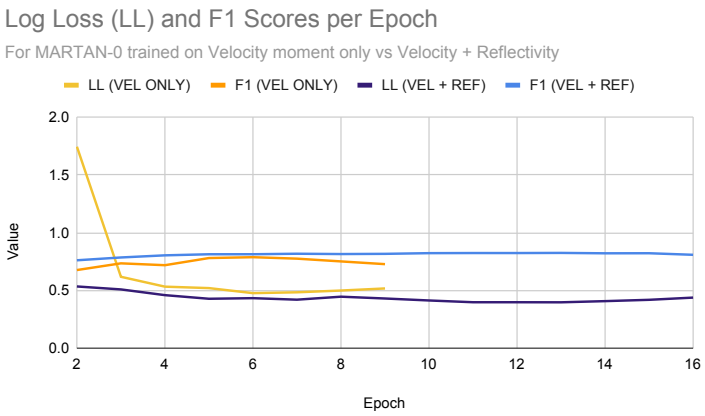


Figure 7. Comparing F1 and Log Loss for mesoscale identification binary classification reveals that including the reflectivity moment to the model significantly increases both classification accuracy and training stability.

MARTAN-0 also performs surprisingly effectively as a binary classifier for tornadic feature detection, despite the very restricted amount of data supplied (see Figure 7). Within the testing set, the model successfully classified a warned or confirmed tornadic sample in 83.11% of cases, and successfully classified the absence of these features in 81.69% of cases. These results indicate that MARTAN-0 can be suitably used as a TVS localiser, or as an acceptable low-overhead initiator model for other, more in-depth radar analysis models.

3.2. MARTAN-1

The initial MARTAN-1 models were trained on a reduced input space centred on each sample's associated warning centroid. These initial results clearly indicate that in this scenario, CoordConv layers hindered the classification performance of the model (see Figure 4). The methodology and reasoning for using CoordConv[9] layers are different in this scenario compared to Veilette et al's [3] original design, as in this implementation, the TVS was assumed to be centred within the input. In practice, however, it's difficult to ensure that the AOI is centred on the input for each sample. MARTAN-0 can attempt to identify an AOI for each sample, however it can be imprecise with this, and cannot be assumed to create an accurate AOI centroid for every positive sample. A strongly supervised, fully labelled dataset that provides accurate mesocyclone centroids could unlock this avenue and would substantially improve MARTAN-0's localisation performance. For this current scenario, the performance of the network with traditional convolutional layers was superior to the alternative coordinate convolutional layers.

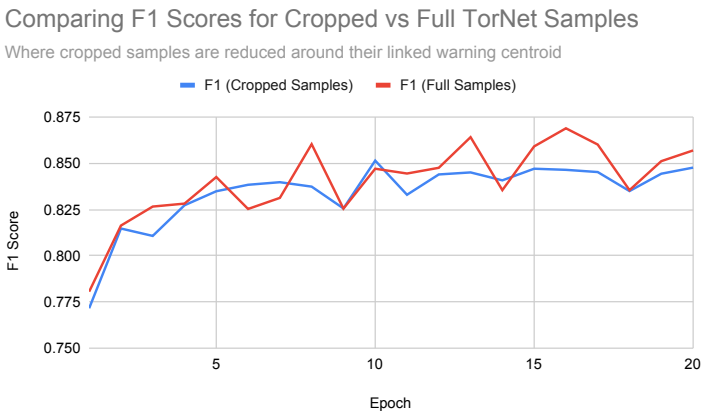


Figure 8. Comparing F1 scores between the full TorNet samples, and samples spatially reduced to 120×120 around their warning's centroid.

The model performs significantly better without reducing the original TorNet sample dimensions. Models using the full 60 degree by 240 range bin resolution outperformed the models that were cropped around their linked warning centroid, and the earlier models cropped around identified MARTAN-0 centroids. It would be difficult to increase the dimensions further, as this could lead to contamination by multiple mesocyclones being included within each sample. A trade-off should be found between including as much information as possible, without risking the size of the extracted radar patch being so large that multiple AOIs are included within the same input space. Future models could explore this equilibrium further, but this model is restricted to the set TorNet sample size.

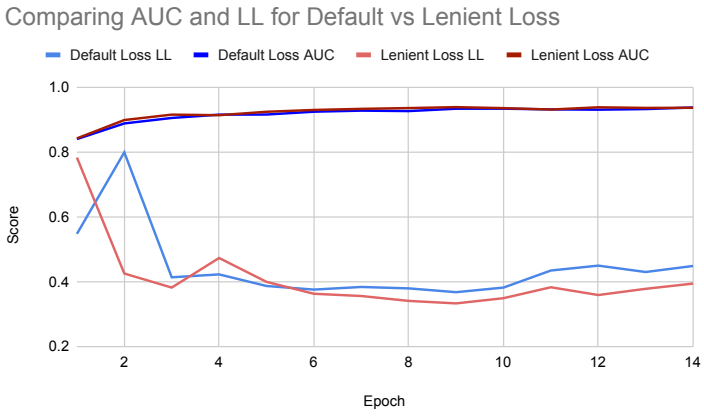


Figure 9. A comparison of default vs lenient loss by looking at the Area Under the Curve and Log Loss throughout the training epochs. Lenient loss provides a significant improvement to log loss.

MARTAN-1 v0.2.0 and v0.3.0 are the two first versions tested on live tornado events. MARTAN-1 v0.2.0 used the default training loss, whereas v0.3.0 used the "lenient" loss, reducing loss by half on warned samples with a positive prediction. This alteration has had a positive impact on identifying positive cases, with a significant improvement in log loss and a slight improvement in AUC in deeper epochs (see Figure 9). The 9th epoch for v0.3.0 was used for live testing, as well as the following performance evaluations (complete table of performance metrics throughout training provided in Appendix A).

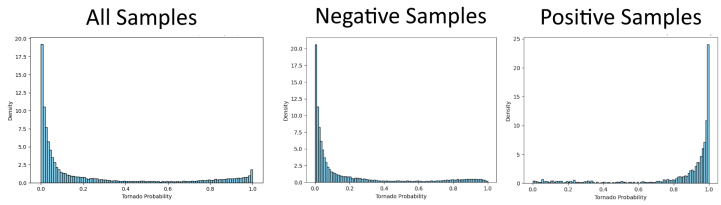


Figure 10. Distribution of predictions across the entire testing set, across negative samples within the testing set, and positive samples within the testing set.

Table 4. Evaluation metrics when discriminating between different sample types

Split	Acc	Prec	Rec	F1	Log Loss	Brier Score	ROC-AUC	Pos Acc	Neg Acc
TOR | WRN+SVR+NONE	0.864	0.855	0.877	0.866	0.333	0.11	0.938	0.877	0.851
TOR+WRN | SVR+NONE	0.708	0.894	0.472	0.618	0.753	0.166	0.846	0.472	0.944
TOR | WRN	0.753	0.702	0.877	0.78	0.581	0.239	0.863	0.877	0.628
TOR | SVR+NONE	0.911	0.94	0.877	0.907	0.23	0.047	0.97	0.877	0.944
TOR | NONE	0.921	0.962	0.877	0.917	0.206	0.035	0.978	0.877	0.965

Table 5. Comparing ROC-AUC performance of a couple of popular tornado detection models with MARTAN-1.

Model	Tornado vs All	Tornado vs Negative	Tornado vs Warned
MARTAN-1	0.938	0.978	0.863
TorNet	0.874	0.921	0.793
TORP	0.89	N/A	N/A

Achieving a ROC-AUC of 0.938 across the entire dataset, MARTAN-1 can consistently differentiate between a sample featuring tornadic features compared to samples that do not depict tornadic features. It correctly classified 88.76% of positive samples ("tornado" labelled samples) and 82.1% of negative samples ("warned", "severe", "negative" labelled samples). By comparing the model's performance at discriminating between "tornado" and "warning" labels and "severe" and "negative" labels, we get an understanding of the impact the complicated "warning" class has on the performance of the model. Remember that "warning" labelled samples depict tornadic features significant enough to elicit a tornado warning from meteorologists, without a tornado being confirmed after the fact.

When treated as a negative label, "warning" samples were only classified correctly in 62.81% of samples. A substantial increase in log loss and a reduction in ROC-AUC in comparisons including "warning" samples reinforces this claim. By removing "warning" samples from consideration, we see a significant increase in classification accuracy across the dataset, and a great increase in model confidence indicated by the substantial reduction in Brier score. By comparing confirmed "tornado" samples against "severe" and "negative" samples, we get an insight into the considerable skill demonstrated by the model when asked to differentiate between clearer positive and negative samples.

A ROC-AUC score of 96.95% and a negative accuracy of 96.52% when discriminating "tornado" against "severe" and "negative" samples indicate that the model can clearly differentiate between samples that feature robust convective precipitation and those that contain a similar representation with added tornadic features. These results clearly indicate that the model can generalise effectively when attempting to estimate tornado likelihood.

3.3. MARTAN-2

Table 6. Classification accuracy by environment intensity label for MARTAN-2

Label	Precision	Recall	F1
None	0.978	0.998	0.988
Weak	0.913	0.576	0.707
Strong	0.77	0.721	0.744

MARTAN-2 is the most experimental of the three models, and so these results should be viewed as a proof of concept rather than a robust, skilful model. Due to the fact that the environment differs more on an event-by-event basis rather than a tornado-by-tornado basis, there exists less opportunities to test the model in live scenarios. Furthermore, tornado events this spring have generally failed to produce discrete tornadic supercells, leading to even fewer higher parameter samples to test the model on. By analysing the evaluation metrics produced by the training process, we can gain an insight into whether the random forest model can generally differentiate between different severities of tornadic environments.

A macro F1 of 0.813 suggests that the model can differentiate well between the different environment types, but the lower F1 score for classifying weak and strong environments tells us that it struggles in separating lower-end tornadic environments with higher-end environments. Separating weak or strong samples from none samples is a far easier task in general, as the descriptor variables supplied (especially high shear, helecity and cape) can create a good understanding on whether tornados are possible in an environment. Differentiating between strong and weak environments is a far more difficult task however, and this is reflected in the results.

Far more testing must be done on live environments before the model can be considered good enough to produce sensible results. More mesoscale environmental variables could be included to create a richer description of the environment, and better data collection is probably required when testing on live events as the current method is limited to an hour-by-hour analysis. The impact of the environmental variables also changes throughout the year, with early spring favouring higher shear due to less available moisture than in late spring-summer, information which the model currently doesn't have access to. Nevertheless, the results during testing on the TorNet dataset indicate the potential for random forests to be suitable for use in this scenario.

3.4. Explainable AI Analysis

The following are four distinct case studies from the test set. The first depicts an obvious tornadic sample that the model performed well on. The second sample depicts some general non-severe thunderstorms on which the model performed well. The third depicts a clipped sample taken from either the close or far limit of the radar that the model struggled with. The last sample depicts a severe-warned non-tornadic sample that the model classified aggressively incorrectly. Each CAM is extracted from the final convolutional layer unique to each moment's head. The sum of activations of each channel are computed using the first-order Taylor expansion on PyTorch's autograd to produce a channel-wise attribution for the prediction [11].

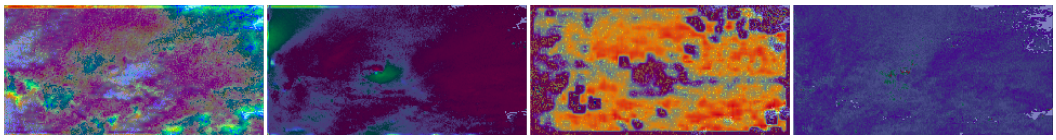


Figure 11. Reflectivity, Velocity, Correlation Coefficient, and Spectrum Width activations for case study 1.

Table 7. Per-channel sum of activations for case study 1. MARTAN-1 prediction: 98.8%.

REF_0	REF_1	VEL_0	VEL_1	CC_0	CC_1	SW_0	SW_1
-0.054	3.52	2.52	0.53	-29.293	0.438	0.375	-0.256

The velocity head clearly shows an area of activation around the TVS, while the reflectivity head shows sporadic activations around the strongest areas of precipitation. The correlation coefficient head shows significant activation in homogenously high corelation regions, suggesting that this head pays more attention to areas where correlation remains consistent, rather than looking for areas where the correlation drops. Spectrum width fails to highlight any areas of activation at all. The bottom-tilt correlation coefficient tensor has a significant negative weighting towards the final prediction, with only highest-tilt reflectivity and lowest-tilt velocity possessing a distinct positive weighting. This suggests that lowest-tilt velocity and highest-tilt reflectivity are weighted the highest when formulating the final prediction.

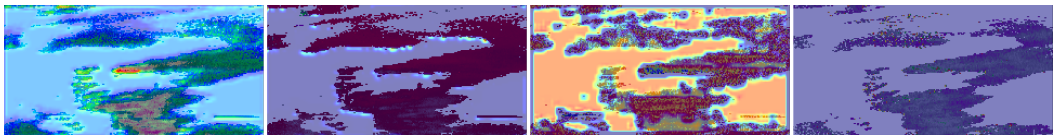


Figure 12. Reflectivity, Velocity, Correlation Coefficient, and Spectrum Width activations for case study 2.

Table 8. Per-channel sum of activations for case study 2. MARTAN-1 prediction: 3.61%.

REF_0	REF_1	VEL_0	VEL_1	CC_0	CC_1	SW_0	SW_1
-1.30	-2.254	0.608	-1.037	-15.14	0.699	-3.99	-2.433

The activations on the reflectivity CAM have clearly highlighted the most substantial cell of the sector. The model appears to pay more attention to the periphery of cells, especially across a steep gradient (I.E strong precipitation dropping to 0 over a short area). Interestingly, the larger mass of storms to the south has been ignored by the reflectivity head, despite having similar reflectivity readings. This suggests that the reflectivity head has learned how to identify good storm structure as a whole, which is one of the key benefits of using a CNN over extracted feature values, where this context is lost. The velocity CAM highlights areas of higher velocity appearing in pixels above those showing negative or no velocity. This indicates that the velocity head is looking for TVSs effectively, but can differentiate between TVSs and general boundaries, since the velocity channel did not advocate for a higher tornado probability despite highlighting many velocity gradients.

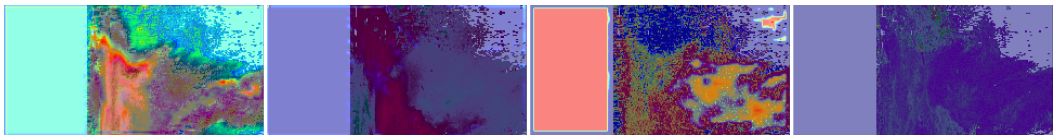


Figure 13. Reflectivity, Velocity, Correlation Coefficient, and Spectrum Width activations for case study 3.

Table 9. Per-channel sum of activations for case study 3. MARTAN-1 prediction: 67.72%.

REF_0	REF_1	VEL_0	VEL_1	CC_0	CC_1	SW_0	SW_1
-0.499	0.378	0.064	1.295	-15.516	-0.668	1.745	-0.98

This sample indicates a clear failure scenario for the model. Many samples within the TorNet dataset have their data clipped if the dimensions of the extracted segment around the sample's centroid reach beyond either the lowest or highest valid range for the radar sweep. The sporadic activations shown along the cut off in the velocity CAM indicate that this is having an impact on the final prediction. Improvements can be made by detecting these clipped samples and removing them from the training set, as the model has clearly learned not to deal with them. This issue does not carry through to operational use, as the extracted live radar data can be clamped to the maximum and minimum range, ensuring the full 120 by 240 input.

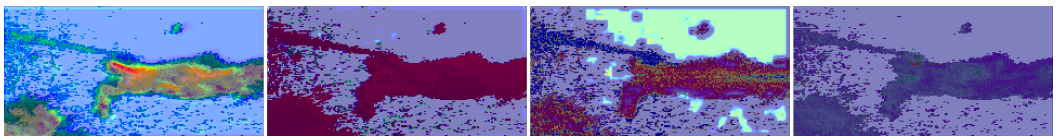


Figure 14. Reflectivity, Velocity, Correlation Coefficient, and Spectrum Width activations for case study 4.

Table 10. Per-channel sum of activations for case study 4. MARTAN-1 prediction: 83.91%.

REF_0	REF_1	VEL_0	VEL_1	CC_0	CC_1	SW_0	SW_1
0.89	-2.222	2.359	2.35	-3.555	0.481	0.195	0.411

A probability of 83.91% represents a significant error by the model when analysing this "negative" labelled sample. Several areas of activations are shown on the velocity CAM around pixels where the range folding has failed causing artificially very steep gradients between pixels. The correlation coefficient input features significant contamination producing large swaths of artificially low correlation throughout the sample. The reflectivity input could also be causing some confusion, due the appearance of a hook feature often associated with tornadic supercells, which is highlighted specifically by the reflectivity CAM. There is perhaps a slight velocity couplet visible in the velocity input, but this appears to be a boundary feature when spatially aligned with the reflectivity input. The contamination, the appearance of general tornadic features, or a combination of several factors could be the source of the erroneously strong prediction.

4. Discussion

MARTAN-1's ability to differentiate between confirmed and warned samples indicate that it has not just learned how to separate tornadic from non-tornadic features, but has also learned how to differentiate between samples that contain tornadic features without producing a tornado, and those that do. By analysing the features flagged in the

CAM analysis of the case studies, the model is successfully utilising the added contextual information provided by processing the entire radar segment to better inform its prediction. This represents a distinct advantage in using CNNs for tornado prediction over the previous predominantly feature value extraction methods. By comparing results with the original TorNet CNN developed by Veilette et al. [3], MARTAN-1 has a lower accuracy of 86.4%, compared to the 95% of TorNet. The primary improvement of MARTAN-1 over TorNet comes from the model's ability to discriminate confirmed vs warned samples, as MARTAN-1 possesses an AUC of 0.863, compared to 0.793. This leads to an overall AUC of 0.938 for MARTAN-1, compared to TorNet's 0.874. It should be noted that because of the extensive dataset refinement described earlier in the paper, there exists some differences between the dataset TorNet was trained on and the dataset MARTAN was trained on.

The current machine-learning model used by NWS meteorologists is TORP (TORnado Probability algorithm), developed and maintained by the NSSL (National Severe Storm Laboratory). TORP is a random forest model that uses extracted feature values and a random forest architecture to form its predictions[1]. MARTAN-1 also possesses a higher AUC than TORP's 0.89, which gives further support to the theory that larger-scale features such as storm presentation matter to predicting tornadogenesis, which is lost when feature values are extracted.

There are several ways to initiate MARTAN-1. As MARTAN-1 requires the input tensor to be in the shape $(1, 2, 120, 240)$, the radar data must be taken from the sweep around a point (X, Y). MARTAN-0 can be used to identify mesocyclones or general AOIs within the sweep, from which the point can be derived using the WSOL method described earlier. Another option could be using the level 3 NEXRAD TVS (Tornado Vortex Signature) product supplied by NOAA[12]. These provide points which have been calculated using an algorithm to potentially contain a tornado signature, which could be extracted and passed to MARTAN-1. Another method could be to treat the model as a "sliding window", where the radar sweep is broken into windows (overlapping to ensure edge features aren't separated), where each window would possess a tornado prediction. For simplicity sake, MARTAN-1 was initiated on the centroids of tornado warning and severe thunderstorm warning polygons, issued by meteorologists.

Some preliminary model biases have been identified from live testing (see Appendix B), the model appears to struggle when analysing warnings issued either extremely close to the radar site, or very far from the radar site. This could be the result of the vastly differing density of the datapoints at close vs long range, or could be the result of the context window not containing the tornadic portion of the scan at extreme close range, due to the centroid of the warning polygon being too far from the AOI to be contained in the extracted radar segment. MARTAN-1 also appears to output high tornado probabilities when faced with a strong wall of wind, such as a QLCS or general gust front. This could be a result of the model over-relying on higher velocity values, or instead the model could have learned that these systems often produce tornados, which is valid in QLCS setups.

Several future improvements can be made to MARTAN-1, more tilts could be supplied to the model, or an increased number of moments. Increasing the number of positive samples in the training dataset could also improve training stability, and help account for the significant class imbalance. MARTAN-2 could potentially be improved by adding more surface analysis variables, and could also benefit from a reduced class imbalance within its training dataset. Accounting for seasonal variability inherent to a few environment variables could also improve model performance.

I hope that this application for deep learning will continue to be improved over the coming years. There exists very limited literature around tornado detection algorithms, especially those that incorporate machine learning components. While the accuracy of

predictions is hamstrung by the current resolution limitations of the measurement instruments, I have no doubt that analysing the evolution of a storm over several time steps could also give a better insight into tornado probability than just analysing a singular moment in time, as is the case with MARTAN. More research can also be conducted into whether deep learning methods such as this can generalise across multiple climates, by incorporating tornadic samples (while undeniably infrequent) collected from doppler radar across multiple continents.

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Data Availability Statement: The original data presented in the study are openly available at GitHub.

Acknowledgments:

- Iowa Environmental Mesonet - Data source [5]
- SPC Tornado Database - Data source [13]
- JJ Helmus and SM Collis, JORS 2016, doi: 10.5334/jors.119 - PyArt radar visualisation library (GitHub)

Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A. MARTAN-1 v0.3.0 training evaluation metrics

v0.3.0	LL	Brier	AUC	ACC	PREC	REC	F1	TP	FP	TN	FN
1	0.7828	0.4031	0.842	0.701	0.632	0.692	0.7629	2327	18577	14621	92
2	0.4252	0.1365	0.8985	0.818	0.8193	0.816	0.8176	1973	5988	27210	446
3	0.3819	0.1402	0.9152	0.8365	0.8153	0.87	0.8418	2105	6542	26656	314
4	0.4731	0.2249	0.9134	0.807	0.7458	0.929	0.828	2248	10451	22747	171
5	0.3995	0.1611	0.924	0.838	0.8018	0.898	0.8472	2172	7375	25823	247
6	0.3628	0.1296	0.9296	0.849	0.8286	0.88	0.8535	2129	6026	27172	290
7	0.3559	0.1221	0.933	0.8545	0.8399	0.876	0.8576	2118	5540	27658	301
8	0.3408	0.0898	0.9354	0.8605	0.8728	0.844	0.8582	2041	4084	29114	378
9	0.3331	0.1101	0.9383	0.864	0.8548	0.877	0.8657	2122	4934	28264	297
10	0.3492	0.1167	0.9352	0.8595	0.946	0.879	0.8622	2126	5316	27882	293
11	0.3829	0.1063	0.9309	0.859	0.8612	0.856	0.8586	2070	4593	28605	349
12	0.3589	0.0928	0.9378	0.8665	0.872	0.859	0.8655	2077	4198	29000	342
13	0.3781	0.105	0.9359	0.865	0.8593	0.873	0.8661	2111	4759	28439	308
14	0.3942	0.0797	0.9363	0.868	0.889	0.841	0.8643	2035	3482	29716	384
15											
16											
17											
18											
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20											

Figure A1. Complete evaluation metrics for each epoch whilest training MARTAN-1 v0.3.0. Epoch 9 was selected for operational use as it produced the best results before overfitting.

Appendix B. MARTAN-1 live performance examples

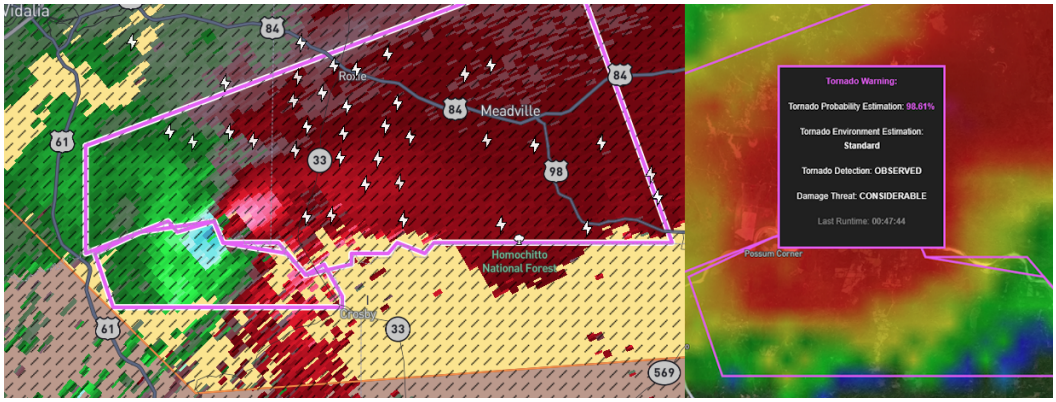


Figure A2. A PDS tornado warning with a very strong mesocyclone. MARTAN-1 gave a very high tornado probability prediction.

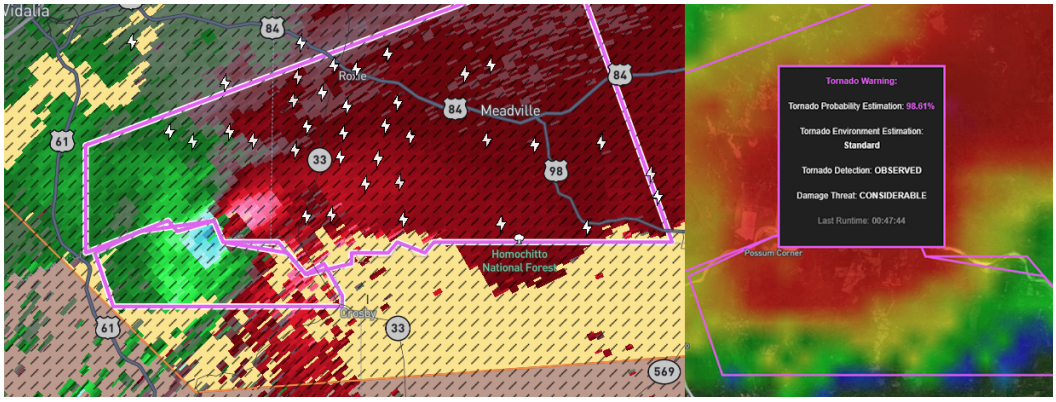


Figure A3. A PDS tornado warning with a very strong mesocyclone. MARTAN-1 gave a very high tornado probability prediction.

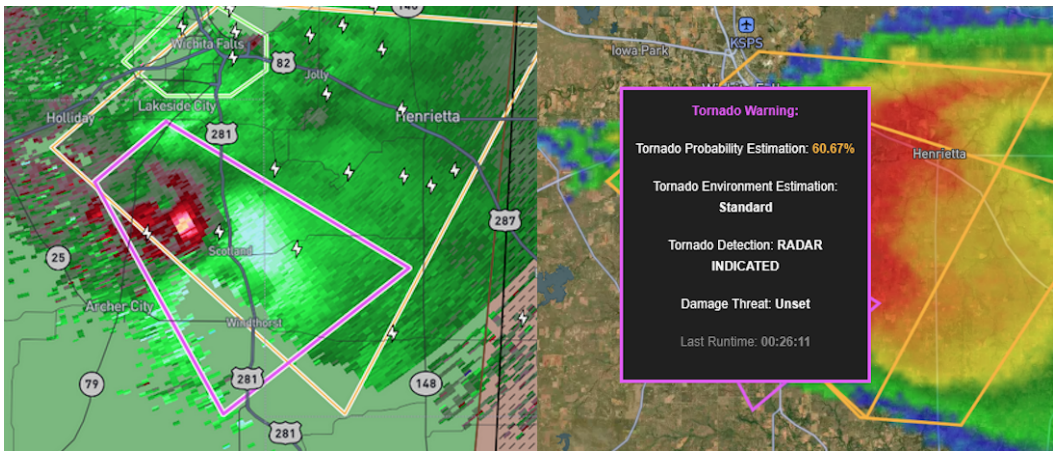


Figure A4. A strong but broad mesocyclone that MARTAN-1 gave a moderate tornado probability prediction.

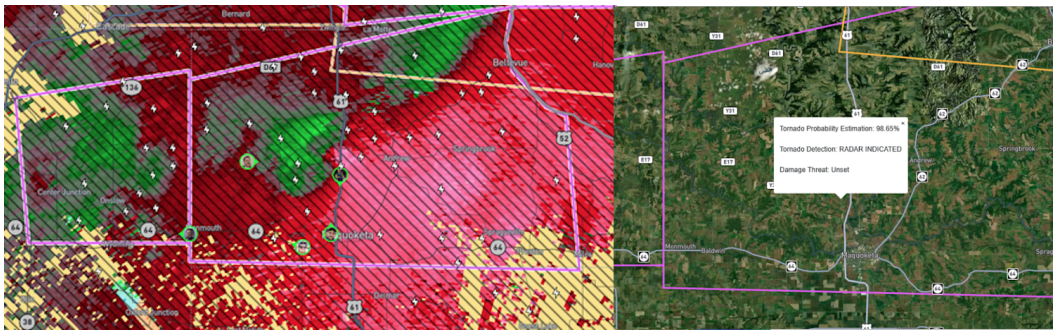


Figure A5. An earlier trial of MARTAN-1, produced a very strong tornado prediction while the warning was still "radar indicated". Warning was upgraded to "observed" soon after.

Appendix C. MARTAN-1 ablation studies

Table A1. Ablation studies with some flavour of data removed for each trial. MARTAN-1 v0.3.0 is the baseline model.

Trial	Converge Epoch	LL	Brier	AUC	F1
Baseline	9	0.3331	0.1101	0.9383	0.8657
-REF	5	0.3869	0.0862	0.9186	0.8199
-VEL	8	0.3953	0.1199	0.9047	0.828
-RHO	13	0.333	0.1058	0.9329	0.8605
-WID	6	0.3699	0.1209	0.9214	0.8439
-RHO, -WID	8	0.3629	0.1203	0.9261	0.8526
-Tilt 1	8	0.346	0.1134	0.9322	0.8583

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